

Functionalities and Tools of OpenAI Agent SDK Not Offered by Ollama

Below is a list of key features and capabilities available in OpenAI's Agent SDK that are not provided (or require significant manual implementation) when using Ollama:

* **Built-in Agent System**

- * OpenAI provides ready-made Agent classes, tool definitions, and function-calling helpers that integrate automatically with their models.

- * Ollama requires you to manually implement all agent logic, routing, parsing, and error handling.

* **Memory**

- * OpenAI supports conversation memory, allowing context and user variables to persist across multiple interactions.

- * Ollama does not have built-in memory; you must manage conversation history manually if needed.

* **Automatic Tool Routing**

- * OpenAI can automatically choose and call the appropriate tool from a provided list.

- * In Ollama, you need to handle routing manually through custom classification logic.

* **Parallel Function Calling**

- * OpenAI allows calling multiple tools in a single step, enabling workflows like classification, summarization, and logging at once.

- * Ollama can only run one tool at a time, requiring manual orchestration for parallel execution.

* **Code Execution (Code Interpreter)**

- * OpenAI includes a built-in code interpreter that can run Python code in the cloud, supporting tasks like math, data analysis, CSV processing, and generating plots.

- * Ollama cannot execute code automatically; any code execution features must be implemented manually.

* **File Handling and Vision**

- * OpenAI supports native file uploads (PDFs, images, CSVs) and includes GPT-4 Vision for direct image analysis.

- * Ollama lacks built-in file upload and vision capabilities; you need to handle OCR and image parsing manually.

* **Structured Function Calling with Validation**

- * OpenAI supports functions with strict JSON schemas, allowing models to automatically produce validated, structured outputs.

- * Ollama can only prompt for JSON extraction without built-in schema validation or enforcement.